2. **Handling Missing Data:**
    - Random forests can handle missing data by using surrogate splits or averaging predictions from trees that use different features.

When to Use Decision Trees and Random Forests

- **Decision Trees:**
    - When interpretability is crucial.
    - For smaller datasets where overfitting is less likely.
- **Random Forests:**
    - For large datasets with complex relationships.
    - When high accuracy and robustness are required.
    - When dealing with noisy or high-dimensional data.

By understanding and applying decision trees and random forests, you can solve a wide range of machine learning problems, from classification to regression, with confidence and precision.

Evaluating Models: Accuracy, Precision, and Recall

Evaluating the performance of a machine learning model is a crucial step in ensuring that it generalizes well to unseen data. Proper evaluation helps identify the strengths and weaknesses of a model and guides improvements for better accuracy and reliability. In this section, we will explore commonly used evaluation metrics such as accuracy, precision, and recall, with a focus on their practical applications and limitations.